XIAO **SHANG**, P.Eng

+1 (647) 389-9165 | xiao.shang@mail.utoronto.ca | Personal website | Google Scholar | LinkedIn | Toronto, Ontario, Canada

Education

University of Toronto — Toronto, ON, Canada
Doctor of Philosophy - Materials Science and Engineering (GPA 4.0/4.0) — *May 2021 – April 2026*

McGill University — Montréal, QC, Canada
Master of Science – Mechanical Engineering (GPA 4.0/4.0) — *2015 –2017*

South China University of Technology — Guangzhou, *China*
Bachelor of Engineering – Mechanical Engineering (GPA 3.8/4.0) — *2011 - 2015*

Research Experience

Lab for extreme mechanics & additive manufacturing, University of Toronto — Toronto, ON, Canada
Graduate Research Assistant (Supervised by Prof. Yu Zou)
Design and additive manufacture of high performing metallic functionally graded materials via microstructure modification — *2021 - Present*

MSM@Harwell Group, University College London — Didcot, UK
Visiting Scholar (Supervised by Prof. Peter D. Lee) — *Dec. 2023, Dec. 2024, Jan. - Feb. 2025*
Investigate magnetic field assisted additive manufacturing in synchrotron X-ray

Brunner Neutrino Lab, McGill University — Montréal, QC, Canada
Research Engineer (Supervised by Prof. Thomas Brunner) — *2018 - 2021*
Provide engineering support to various nuclear physics experiments

Architected materials and advanced structures group, McGill University — Montréal, QC, Canada
Graduate Research Assistant (Supervised by Prof. Damiano Pasini) — *2015 – 2017*
Durable bistable auxetics made of rigid solids

Micro engineering, dynamics and automation lab, University of Calgary — Calgary, Canada
Mitacs Globalink research intern (Supervised by Prof. Simon Park) — *June - Sept. 2014*
Investigate the effects of sapphire tooling on the machining process

Publications and Awards

Selected publications (Full list at **Google Scholar**)

1. **Shang, X.**, Talbot, A., Li, E., Wen, H., Lyu, T., Zhang, J., & Zou, Y. (2025). Accurate inverse process optimization framework in laser directed energy deposition. *Additive Manufacturing*, *102*, 104736. *2025*
2. **Shang, X.**, Liu, Z., Zhang, J., Lyu, T., & Zou, Y. (2023). Tailoring the mechanical properties of 3D microstructures: A deep learning and genetic algorithm inverse optimization framework. *Materials Today, 70, 71-81.* *2023*
3. **Shang, X.**, Liu, L., Rafsanjani, A., & Pasini, D. (2018). Durable bistable auxetics made of rigid solids. *Journal of Materials Research, 33*(3), 300-308. *2018*
4. **Shang, X.**, Evelyn L., Soumya S. D., Ajay T., & Zou, Y. (2025). Additive manufacturing of multi-metallic materials to achieve multi-functionality. Under review at *Science Advances*. *2025*
5. **Shang, X.**, Dash, S. S., Shao, C., Lang, L., … & Zou, Y. (2025). Co-deformation failure mechanisms in directed energy deposited bi-metallic steels and a pathway to improve their ductility and toughness. Submitted to *Acta Materialia.* *2025*
6. Zhang, J.*, **Shang, X.***, … & Zou, Y. (2025). Mechanism and Quantification of Melt Pool Morphology Evolution in Single-Track Fabrication by Laser Direct Energy Deposition. Under review at *Journal of Manufacturing Processes*. * indicates co-first authors. *2025*
7. Talbot, A., **Shang, X.**, … & Zou, Y. (2025). Laser remelting of a CrMnFeCoNi high entropy alloy: Effect of energy density on elemental segregation. *Advanced Engineering Materials*, 2501194. *2025*
8. Dash, S. S., Fernandes, R. C., **Shang, X.**, Zou, Y., ... & Chen, D. (2025). Fatigue and deformation mechanisms of ultrasonic spot-welded dissimilar joints of a magnesium alloy to a clad aluminum alloy. *Journal of Magnesium and Alloys*. *2025*
9. Dash, S. S., Lang, L., Agyapong, J., Yang, S., **Shang, X.**, ... & Zou, Y. (2025). Hetero-deformation and fracture of a near-eutectic high entropy alloy under dynamic impact. *Scripta Materialia*, 260, 116593. *2025*
10. Wang, W., Choi, H. S., Dash, S. S., Lyu, T., **Shang, X.**, & Zou, Y. (2025). Self-generating lubricious oxides for friction reduction in FeCoNiMo and CrCoNiMo from room temperature to 1000° C. *Journal of Alloys and Compounds*, 1022, 180003. *2025*
11. Zhang, J., Jiang, R., Li, K., Chen, P., Bi, S., **Shang, X.**, ... & Zou, Y. (2025). Accurate and efficient predictions of keyhole dynamics in laser materials processing using machine learning-aided simulations. *International Journal of Heat and Mass Transfer*, 250, 127279. *2025*
12. Chen, H., Lang, L., **Shang, X.**, Dash, S. S., He, Y., King, G., & Zou, Y. (2024). Anisotropic co-deformation behavior of nanolamellar structures in additively manufactured eutectic high entropy alloys. *Acta Materialia*, *271*, 119885. *2024*
13. Lyu, T., Keshavarz, M. K., Patel, S., Haché, M. J., Cheng, C., **Shang, X.**, ... & Zou, Y. (2024). Melting mode driven tuning of local microstructure and hardness in laser powder bed fusion of 18Ni-300 maraging steel. *Materialia*, 35, 102113. *2024*
14. Adhikari, G., ... **Shang. X.**, et. al (2021). nEXO: Neutrinoless double beta decay search beyond 10^{28} year half-life sensitivity. (2021). *Journal of Physics G: Nuclear and Particle Physics*, 49(1), 015104. *2021*
15. Yin, Z., ... **Shang. X.**, … (2021). Young martlets: Exploring the world of academia and beyond. *Matter*, *4*(5), 1434-1436. *2021*

Patents

"Bistable Auxetics" (US20170362414). U.S. Patent and Trademark Office. *2020*

"Modular Ground-Stowable Chairs and Desks" (ZL2014103573363). China National Intellectual Property Administration. *2016*

Selected Awards

Joseph R. Benedetto Scholarship, provincial award ($1,000) *2025*

MITACS Globalink Research Award, international award ($6,000) *2024*

NSERC Michael Smith Foreign Study Supplements, national award ($6,000) *2024*

NSERC Canada Graduate Scholarships – Doctoral (CGS D), national award ($120,000) *2023*

Ontario Graduate Scholarship, provincial award, twice ($30,000) *2021 and 2022*

MITACS Globalink Graduate Fellowship, international award ($15,000) *2015*

MITACS Globalink Research Intern Scholarship, international award ($7,500) *2014*

Chinese National Encouragement Scholarship, national award (¥5,000, top 2%) *2014*

Chinese National Scholarship, national award (¥8,000, top 1%) *2013*

Teaching

CIV1199: Joining, Welding and Additive Manufacturing
University of Toronto | Teaching Assistant *Fall 2025*

MSE403 – Data Sciences and Analytics for Materials Engineers
University of Toronto | Teaching Assistant *Winter 2023, Winter 2024*

MSE1065 – Application of Artificial Intelligence in Materials Design
University of Toronto | Teaching Assistant *Winter 2021, Fall 2022, Fall 2023*

ME243 - Manufacturing and CAD Modeling
Shijiazhuang Tiedao University | Guest Lecturer *Spring 2024*

Subatomic Physics Summer Lecture Series
McGill University | Guest Lecturer *2020-2022*

Voluntary Service

The Minerals, Metals & Materials Society (TMS)

Website | Member of the Solidification Committee *2025 - Present*

- Engage in the program development of the committee. Help shape the conference agenda, suggest panels and special topics. Contribute to long-term planning of the committee's role in the conference.

Materials Science and Engineering Graduate Students' Association

University of Toronto | President *2021– 2022*

- Responsible for the direction and actions of the association and represent our graduate students at the department, faculty, and university level councils.

The Martlets Society

Website | Organizer and Editor-in-chief *2020 - Present*

- The Martlets Society is a non-profit organization funded and run by early career researchers. We invite researchers from top Universities such as MIT, Oxford University, U of T etc.to give talks on their research topics and career advice.